

1. Sea la función Inmitancia:

$$F(s) = \frac{s^4 + 4s^2 + 3}{s(s^2 + 2)}$$

Se pide hallar la topología circuital y los valores de los componentes

- a) Síntesis de $Z(s)$ mediante el método de Foster en su versión serie
- b) Síntesis de $Y(s)$ mediante el método de Foster en su versión derivación
- c) Síntesis de $Z(s)$ mediante el método de Cauer remoción en infinito
- d) Síntesis de $Z(s)$ mediante el método de Cauer remoción en DC
- e) Simular $Z(s)$ en LTSpice, observando los cambios de fase y la ley de variación (pendiente) de la impedancia del dipolo.

2) Foster Impedancia

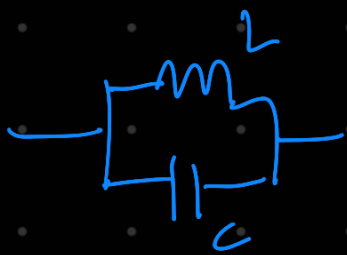
$$Z(s) = k_{\infty} s + \sum \frac{2k_i s}{s^2 + \omega_i^2} + \frac{k_0}{s}$$

$$Z(s) = \frac{s^4 + 4s^2 + 3}{s(s^2 + 2)} = \frac{3/2}{s} + \frac{1/2}{s^2 + 2} + s$$

$$k_0 \rightarrow Z(s) \cdot s \Big|_{s \rightarrow 0} = \frac{0 + 0 + 3}{0 + 2} = 3/2$$

$$2k_i \rightarrow Z(s) \cdot \frac{(s^2 + 2)}{s} \Big|_{s^2 \rightarrow -2} = \frac{4 + 4(-2) + 3}{-2} = \frac{-1}{-2} = \frac{1}{2}$$

$$k_{\infty} \rightarrow Z(s) \frac{1}{s} \Big|_{s \rightarrow \infty} = \frac{\cancel{s^4} \left(1 + \frac{4}{s^2} + \frac{3}{s^4} \right)}{\cancel{s^4} \left(1 + \frac{2}{s^2} \right)} = 1$$



$$Z_{\text{range}} = \left(\frac{1}{sL} + sC \right)^{-1}$$

$$= \frac{sL}{(1 + s^2 CL)}$$

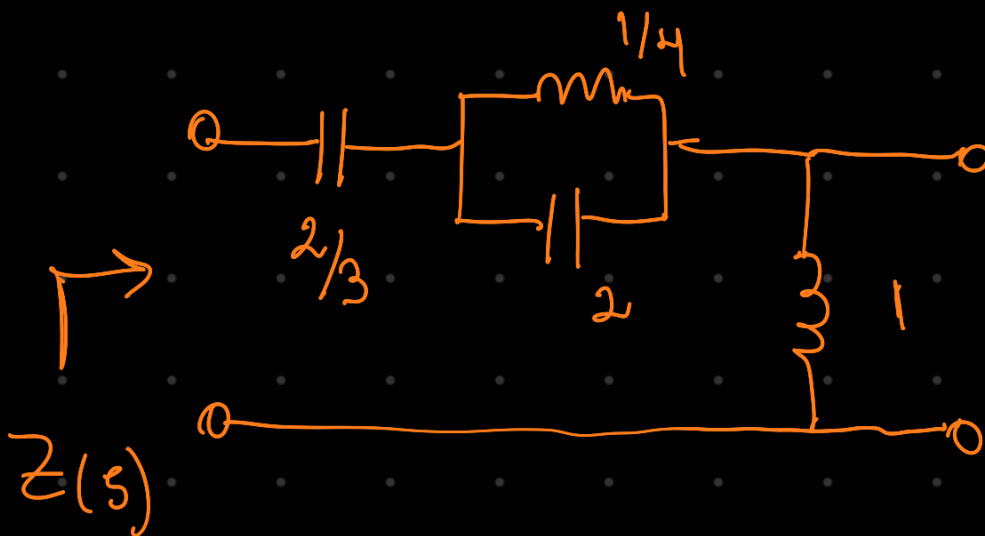
$$\frac{1}{Lc} = 2 \rightarrow Lc = \frac{1}{2} / C$$

$$L = \frac{1}{2} / C$$

$$2\text{Ki} = \frac{1}{C} \wedge L = \frac{2\text{Ki}}{2}$$

$$\frac{5K}{sL \left(s^2 + \frac{1}{LC} \right)}$$

$$C = 2 \quad L = 1/4$$



Verificación

$$Z(s) = \frac{1}{s \frac{2}{3}} + \frac{s \cdot \frac{1}{2}}{s^2 + \frac{1}{2 \cdot \frac{1}{4}}} + s = \frac{3/2}{s} + \frac{s \frac{1}{2}}{s^2 + 2} + s$$

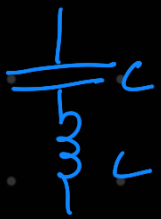
$$\frac{(s^2 + 2) \frac{3}{2} + s \frac{1}{2} s + s^2 (s^2 + 2)}{s (s^2 + 2)}$$

$$\frac{\frac{3s^2+3}{2} + \frac{s^2}{2} + s^4 + 2s^2}{s(s^2+2)} = \frac{s^4 + 4s^2 + 3}{s(s^2+2)}$$

b) Foster Admittance

$$Y(s) = \frac{s^4 + 4s^2 + 3}{s(s^2+2)} = \frac{k_0}{s} + \frac{2k_i s}{s^2+2} + k_\infty s$$

$$k_0 = 3/2 \quad k_\infty = 1 \quad 2k_i = 1/2$$



$$Y(s) = \left(\frac{1}{sL} + sC \right)^{-1} = \left(\frac{1 + s^2 CL}{sC} \right)^{-1}$$

$$Y(s) = \frac{sC}{s^2 + \frac{1}{LC}}$$

$$L = \frac{1}{2k_i} = 2$$

$$\frac{1}{LC} = 2 \rightarrow \frac{1}{2} = LC$$

$$C = \frac{1}{2} \cdot 2k_i = \frac{1}{4}$$



Verifizierung

c) Cover en Inf

$$Z(s) = \frac{s^4 + 4s^2 + 3}{s^3 + 2s}$$

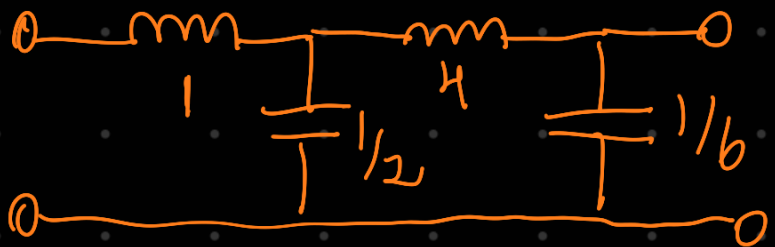
$$\begin{array}{r} s^4 + 4s^2 + 3 \quad | \quad s^3 + 2s \\ - s^4 + 2s^2 \quad \quad \quad s \quad \quad \quad \text{---m---} \\ \hline \end{array}$$

$$\begin{array}{r} 5^3 + 25 \quad \big| \quad 25^2 + 3 \\ - \quad 5^3 + \frac{3}{2}5 \quad \frac{1}{2}5 \quad \frac{1}{2} \frac{1}{2} \\ \hline \end{array}$$

$$\begin{array}{r} 25^2 + 3 \quad \left| \frac{1}{2} 5 \right. \\ - 25^2 + 0 \\ \hline 1512 \end{array} \quad \begin{array}{l} 45 \\ \text{---} \\ 4 \end{array}$$

$$\begin{array}{r} \frac{1}{2} \cdot \frac{1}{3} = \frac{1}{6} \end{array}$$

0/



d) Cover en DC

$$Z(s) = \frac{s^4 + 4s^2 + 3}{s^3 + 2s}$$

$$\begin{array}{r} 3 + 4s^2 + s^4 \overline{) 2s + s^3} \\ \underline{3 + \frac{3}{2}s^2 + 0} \quad \frac{1}{5} \cdot \frac{1}{2} \cdot 3 \\ \frac{5}{2}s^2 + s^4 \\ \underline{\frac{5}{2}s^2 + 0} \quad \frac{4}{5} \quad \frac{1}{5} \quad \left\{ \frac{5}{4} \right\} \\ s^4 \end{array}$$

2/3

$$\begin{array}{r} \frac{5}{2}s^2 + s^4 \overline{) \frac{1}{5}s^3} \\ \underline{\frac{5}{2}s^2} \quad \frac{1}{5} \cdot \frac{25}{2} \\ \frac{1}{5}s^3 \end{array}$$

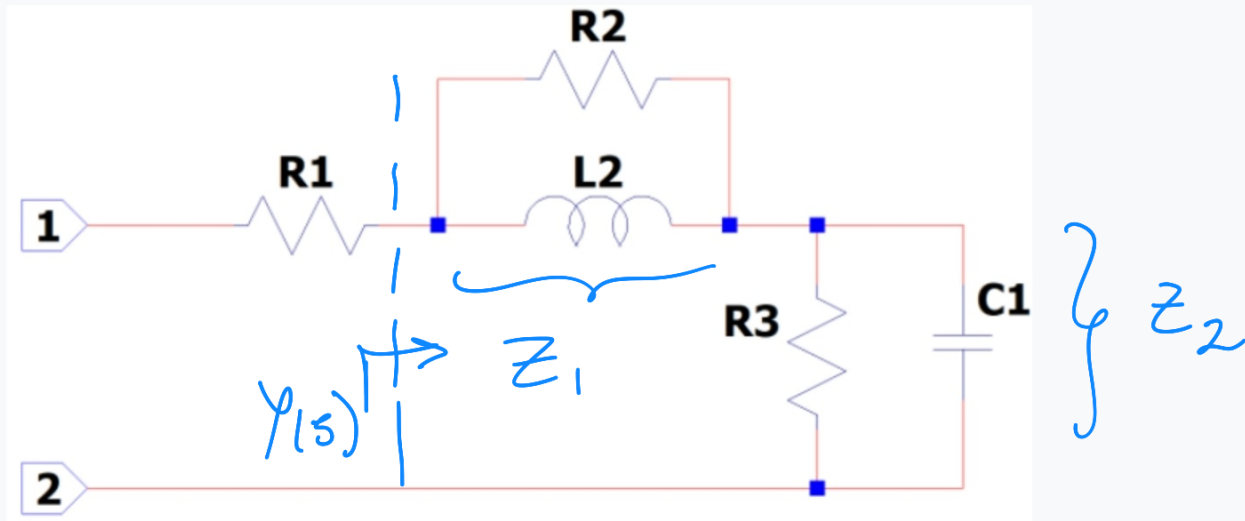
2/25

$$\begin{array}{r} \frac{1}{5}s^3 \overline{) s^4} \\ \underline{\frac{1}{5}s^3} \quad \frac{1}{5} \quad \frac{1}{5} \\ \frac{1}{5}s^3 \\ \underline{\frac{1}{5}s^3} \\ 0 \end{array}$$

5

2. Encuentre el valor de los elementos que integran el siguiente circuito y que satisfacen la función impedancia propuesta:

$$Z(s) = \frac{s^2 + 6s + 8}{s^2 + 6s + 5}$$



$$Z(s) = \frac{s^2 + 6s + 8}{s^2 + 6s + 5} = R_1 + Z_1 + Z_2$$

$$\frac{s^2 + 6s + 8}{s^2 + 6s + 5} \quad \left| \frac{s^2 + 6s + 5}{s^2 + 6s + 5} \right.$$

1

$$s^2 + 6s + 5 \quad \left| \quad 3 \right.$$

$$Y(s) = \frac{s^2 + 6s + 5}{3} \rightarrow Z(s) = \frac{3}{s^2 + 6s + 5}$$

$$Z(s) = 1 + \frac{3}{s^2 + 6s + 5}$$

$$Z(s) = 1 + \frac{3}{(s+1)(s+5)}$$

$$\frac{3}{(s+1)(s+5)} = \frac{k_1}{(s+1)} + \frac{k_2 s}{s+5} = \frac{3/4}{s+1} + \frac{-3/4}{s+5}$$



$$Z(s) = \left(\frac{1}{R} + \frac{1}{sL} \right)^{-1} \rightarrow \frac{sR}{s + \frac{R}{L}}$$



$$Z(s) = \left(\frac{1}{R} + sC \right)^{-1} = \frac{1/C}{s + \frac{1}{RC}}$$

$$-\frac{A}{s+B} + \frac{A}{B} - \frac{A}{B} = \frac{A/B^3}{s+B} - \frac{A}{B} \quad \left. \vphantom{\frac{A}{B}} \right\} \text{Predicted}$$

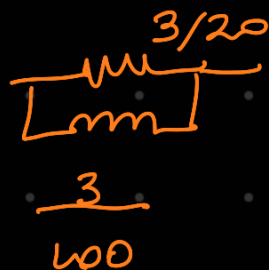
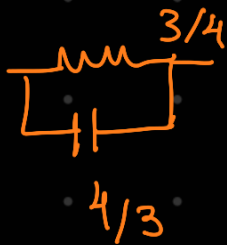
$$-\frac{\overset{A}{3/4}}{s+5} + \frac{3/4}{\underset{B}{5}} - \frac{3/4}{5} = \frac{\frac{3}{20} s}{s+20} \quad \left(-\frac{3}{20} \right)$$

$$\rightarrow R' = 1 - 3/20$$

Se resta el resistor

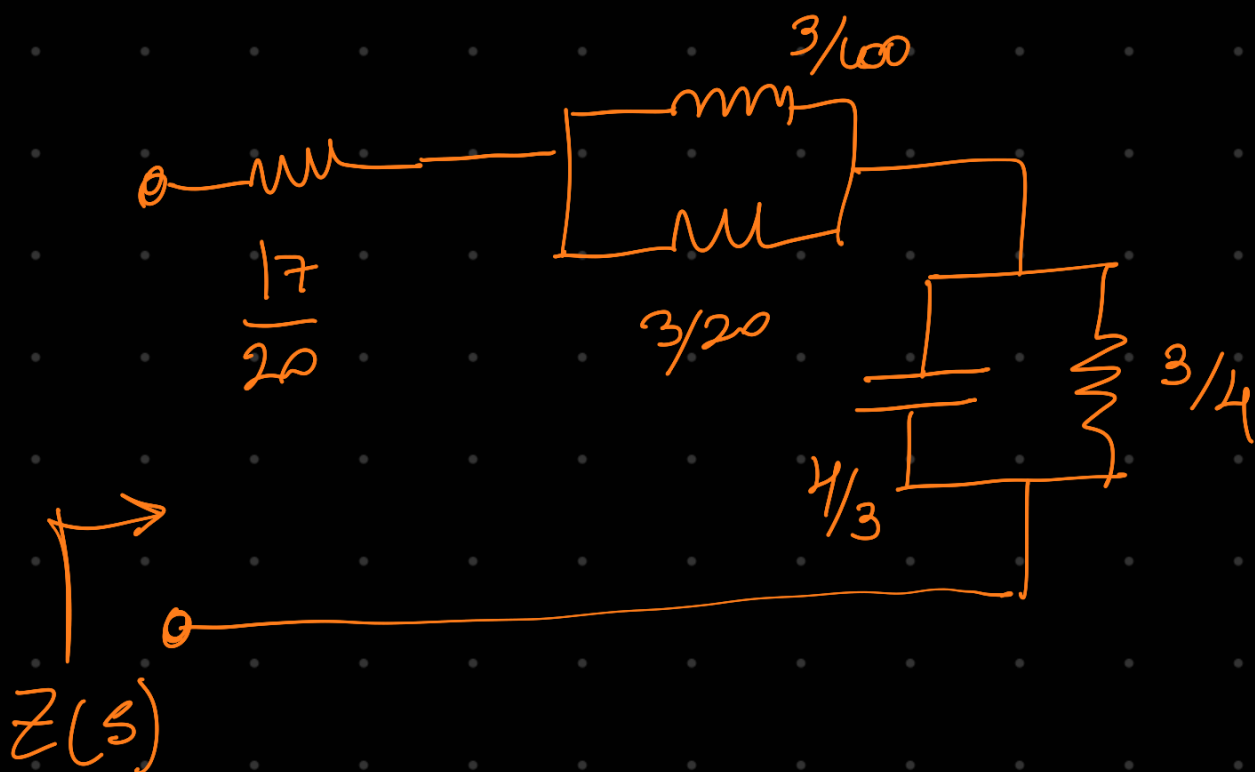
$$Z(s) = \underbrace{\left(1 - \frac{3}{20}\right)}_{17/20} + \frac{\frac{3}{4}}{s+1} + \frac{s - \frac{3}{20}}{s+5}$$

$\frac{17}{20}$
—



$$s = \frac{R}{L}$$

$$L = \frac{R}{5}$$

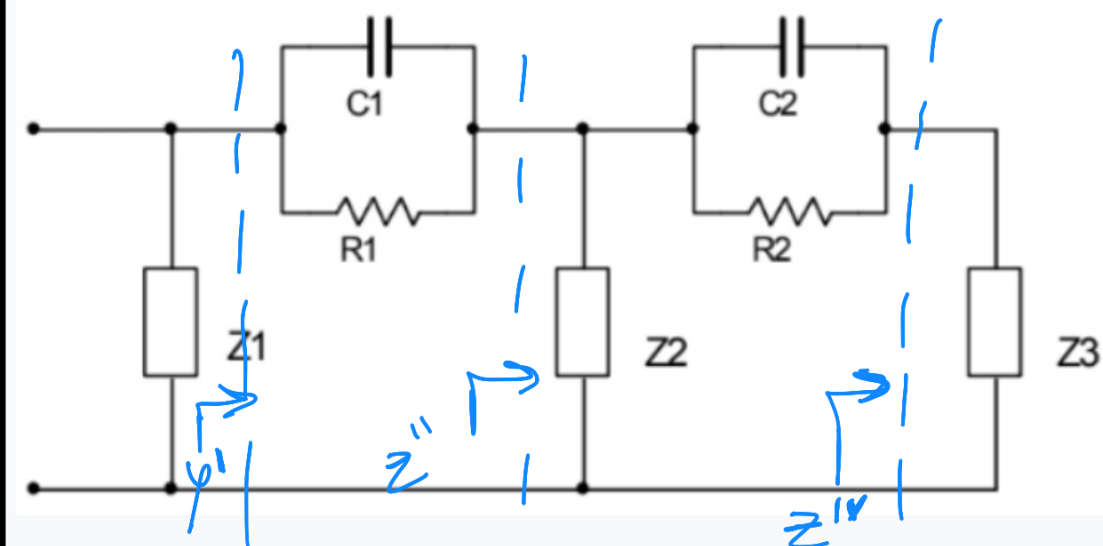


3. Halle el valor de los elementos que integran el siguiente dipolo sabiendo que está caracterizado por las constantes de tiempo indicadas y debe satisfacer simultáneamente el modelo matemático propuesto.

$$Z(s) = \frac{s^2 + 13s + 32}{3s^2 + 27s + 44}$$

$$R_1 \cdot C_1 = 1 \rightarrow \frac{1}{RC} = 1$$

$$R_2 \cdot C_2 = \frac{1}{4} \rightarrow \frac{1}{RC} = 4$$



Hay que setear un cero en (-1) Para luego remover en z

$$Y(s) = \frac{1}{Z(s)} = \frac{3s^2 + 27s + 44}{s^2 + 13s + 32}$$



$$Y' = Y(s) - Y(1) = \frac{3s^2 + 27s + 44}{s^2 + 13s + 32} - \frac{20}{20}$$

Remoción parcial

$$3s^2 + 27s + 44 - (s^2 + 13s + 32)$$

$$s^2 + 13s + 32$$

$$y' = \frac{2s^2 + 14s + 12}{s^2 + 13s + 32} \Rightarrow z' = \frac{s^2 + 13s + 32}{2s^2 + 14s + 12}$$

$2(1+s)(6+s)$

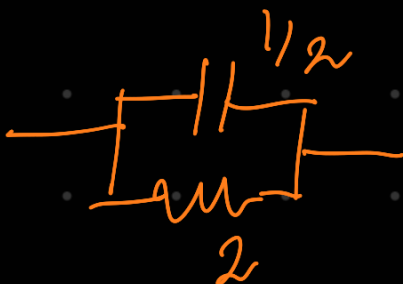
"Remover" el polo $(s+1)$

$$z'' = \frac{s^2 + 13s + 32}{2s^2 + 14s + 12} - \frac{k_1}{s+1}$$

z'

$$k_1 = \frac{1 - 13 + 32}{2(6-1)} = \frac{20}{10} = 2$$

$$C = 1/2 \quad R = 2$$



$$Z = \left(sC + \frac{1}{R} \right)^{-1}$$

$$\frac{1/C}{s + \frac{1}{RC}}$$

$$z'' = \frac{s^2 + 13s + 32}{2(1+s)(6+s)} - 4(6+s)$$

$$\frac{s^2 + 13s + 32 - 24 - 4s}{2(1+s)(6+s)} = \frac{s^2 + 9s + 8}{2(1+s)(6+s)}$$

$$z'' = \frac{\cancel{(1+s)}(8+s)}{2\cancel{(1+s)}(6+s)} = \frac{8+s}{6+s} \cdot \frac{1}{2}$$

Ahora requiero un cero en (-4) Para
 luego remover en z

$$y'' = \frac{(6+s)2}{(8+s)} \Rightarrow y''' = \frac{6+s}{8+s} \cdot 2 - y''(-4)$$

$$y''' = \frac{6+s}{8+s} 2 - \boxed{1} = \frac{(6+s)2 - (8+s)}{8+s}$$

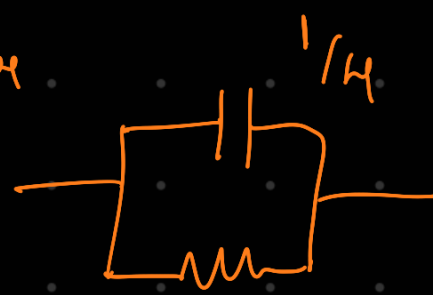
Remoción
 Parcial $\frac{1}{s+1}$

$$Z''' = \frac{8+s}{4+s} \quad \text{Remuevo el tanque}$$

$(4+s)$ Polo en -4

$$Z^{IV} = \frac{8+s}{4+s} - \frac{K_H}{4+s} = \frac{8+s}{4+s} - \frac{4}{4+s}$$

K_H



$$C = \frac{1}{4} \quad \frac{1}{BC} = 4 \rightarrow \frac{4}{B} = 4 \quad B = 1$$

$$Z^{IV} = \frac{8+s-4}{4+s} = \frac{s+4}{s+4} = 1 \Rightarrow Y^{IV} = \frac{1}{Z^{IV}} = 1$$

